

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Classified Practice

Complex Numbers

Q001 **Written****State the real and imaginary parts of $3 - 2i$.**

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

Classified Practice

Complex Numbers

Q003 **Written****State the real and imaginary parts of -8 .**

Q004 Written**State the real and imaginary parts of $2i$.**

Classified Practice

Complex Numbers

Q005 Written

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

Q006 Written**State the real and imaginary parts of $-3 + 5i$.**

Classified Practice

Complex Numbers

Q007 Written

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

Q008 Written**State the real and imaginary parts of 1.**

Classified Practice

Complex Numbers

Q009 Written

State the real and imaginary parts of $3i$.

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0$, x, y real.

Classified Practice

Complex Numbers

Q011 Written

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

Q012 Written**State the real and imaginary parts of $2 + 5i$.**

Classified Practice

Complex Numbers

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Q014 Written**State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.**

Classified Practice

Complex Numbers

Q015 Written

State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.

Q016 Written**State the real and imaginary parts of -4 .**

Classified Practice

Complex Numbers

Q017 **Written****State the real and imaginary parts of $1/4$.**

Q018 Written**State the real and imaginary parts of $\sqrt{3}i$.**

Classified Practice

Complex Numbers

Q019 Written

State the real and imaginary parts of $-i$.

Q020 Written**Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.**

Classified Practice

Complex Numbers

Q021 Written

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.

Q022 Written**Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.**

Classified Practice

Complex Numbers

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.

Q024 Written**Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.**

Classified Practice

Complex Numbers

Q025 Written

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

Q026 Written**Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q027 Written

Calculate $(i - 2)^2$ and give the answer in standard form.

Q028 Written**Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q033 Written

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.

Q034 Written

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q035 Written

Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.

Q036 Written

Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q037 Written

Calculate $(\frac{2}{5} - \frac{1}{2}i) - (\frac{1}{3} + \frac{3}{7}i)$ and give the answer in standard form.

Q038 Written**Calculate $(2i - 1)^2$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q039 **Written****Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.**

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Classified Practice

Complex Numbers

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Q042 Written**Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.**

Classified Practice

Complex Numbers

Q043 Written

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Q044 Written**Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.**

Classified Practice

Complex Numbers

Q045 Written

State the complex conjugate of $2 + 3i$.

Q046 Written**State the complex conjugate of $7 - i\sqrt{11}$.**

Classified Practice

Complex Numbers

Q047 **Written****State the complex conjugate of 6.**

Q048 Written**State the complex conjugate of $-5i$.**

Classified Practice

Complex Numbers

Q049 Written

Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.

Q050 Written**Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q051 Written

State the complex conjugate of $1 - 4i$.

Q052 Written**State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.**

Classified Practice

Complex Numbers

Q053 Written

State the complex conjugate of 2.

Q054 Written**State the complex conjugate of $-8i$.**

Classified Practice

Complex Numbers

Q055 **Written****Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.**

Q056 Written**Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q057 Written

State the complex conjugate of $3 + 4i$.

Q058 Written**State the complex conjugate of $9 + 8i$.**

Classified Practice

Complex Numbers

Q059 Written

State the complex conjugate of $8 + i\sqrt{3}$.

Q060 Written**State the complex conjugate of 5.**

Classified Practice

Complex Numbers

Q061 Written

State the complex conjugate of -15 .

Q062 Written**State the complex conjugate of $\sqrt{7}$.**

Classified Practice

Complex Numbers

Q063 Written

State the complex conjugate of i .

Q064 Written**State the complex conjugate of $-50i$.**

Classified Practice

Complex Numbers

Q065 Written

State the complex conjugate of $i\sqrt{6}$.

Q066 Written**Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q067 **Written****Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.**

Q068 Written

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.

Classified Practice

Complex Numbers

Q069 Written

Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.

Q070 Written**Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q071 Written

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.

Q072 Written**Calculate $1/i$ and give the answer in the form $a+bi$.**

Classified Practice

Complex Numbers

Q073 Written

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

Q074 Written

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)} \right)^{2027}$.

Classified Practice

Complex Numbers

Q075 Written

Express the square roots of -6 in terms of i .

Q076 Written**Express $\sqrt{-12}\sqrt{-6}$ in terms of i.**

Classified Practice

Complex Numbers

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .

Q078 Written**Express $\sqrt{-7}$ in terms of i .**

Classified Practice

Complex Numbers

Q079 Written

Express $\sqrt{-27}$ in terms of i .

Q080 Written**Express $-\sqrt{-12}$ in terms of i .**

Classified Practice

Complex Numbers

Q081 **Written****Express the square roots of -12 in terms of i .**

Q082 Written**Express $\sqrt{-3}\sqrt{-6}$ in terms of i .**

Classified Practice

Complex Numbers

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Q084 Written**Express $(3 + \sqrt{-2})^2$ in terms of i .**

Classified Practice

Complex Numbers

Q085 Written

Express $\sqrt{-2}$ in terms of i .

Q086 Written**Express $\sqrt{-9}$ in terms of i .**

Classified Practice

Complex Numbers

Q087 Written

Express $\sqrt{-45}$ in terms of i .

Q088 Written**Express $-\sqrt{-24}$ in terms of i .**

Classified Practice

Complex Numbers

Q089 Written

Express $-\sqrt{-36}$ in terms of i .

Q090 Written**Express the square roots of -2 in terms of i .**

Classified Practice

Complex Numbers

Q091 Written

Express the square roots of -18 in terms of i .

Q092 Written**Express the square roots of -16 in terms of i .**

Classified Practice

Complex Numbers

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .

Q094 Written**Express $\sqrt{-28}\sqrt{-35}$ in terms of i .**

Classified Practice

Complex Numbers

Q095 Written

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .

Q096 Written**Express $\sqrt{27}/\sqrt{-12}$ in terms of i .**

Classified Practice

Complex Numbers

Q097 Written

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .

Q098 Written**Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .**

Classified Practice

Complex Numbers

Q099 Written

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .

Q100 Written

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .

Classified Practice

Complex Numbers

Q101 **Written****Solve over the complex numbers:** $(2x - 8)^2 = -10$.

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (2, 4)

B (3, 2)

C (2, 3)

D (3, 3)

Classified Practice

Complex Numbers

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-3, 5)$

B $(-4, 6)$

C $(5, -4)$

D $(-4, 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (8, -2)

B (7, -1)

C (-2, 7)

D (7, -2)

Classified Practice

Complex Numbers

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{\sqrt{2}}{3}, \frac{1}{3}\right)$

B $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$

C $\left(\frac{4}{3}, -\frac{\sqrt{2}}{3}\right)$

D $\left(\frac{1}{3}, 1 - \frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6 , select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-5, 0)$

B $(-6, 1)$

C $(0, -6)$

D $(-6, 0)$

Classified Practice

Complex Numbers

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A (0, 4)

B (1, 4)

C (0, 5)

D (4, 0)

Q108 MCQ**For $-i$, select the ordered pair (real part, imaginary part):****Choose the correct option.****A** $(-1, 0)$ **B** $(0, -1)$ **C** $(1, -1)$ **D** $(0, 0)$

Classified Practice

Complex Numbers

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 + \sqrt{5})$

B $(\sqrt{5}, 0)$

C $(0, \sqrt{5})$

D $(1, \sqrt{5})$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(\sqrt{3}, \frac{3}{2})$

B $(\frac{3}{2}, \sqrt{3})$

C $(\frac{5}{2}, \sqrt{3})$

D $(\frac{3}{2}, 1 + \sqrt{3})$

Classified Practice

Complex Numbers

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):**Choose the correct option.**

A $\left(\frac{3}{4}, -2\right)$

B $\left(-\frac{1}{4}, -1\right)$

C $\left(-2, -\frac{1}{4}\right)$

D $\left(-\frac{1}{4}, -2\right)$

Q112 MCQ**For $5/3$, select the ordered pair (real part, imaginary part):****Choose the correct option.**

A $\left(\frac{8}{3}, 0\right)$

B $\left(\frac{5}{3}, 1\right)$

C $\left(0, \frac{5}{3}\right)$

D $\left(\frac{5}{3}, 0\right)$

Classified Practice

Complex Numbers

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 - \sqrt{7})$

B $(-\sqrt{7}, 0)$

C $(0, -\sqrt{7})$

D $(1, -\sqrt{7})$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-8, \sqrt{2})$

B $(-9, 1 + \sqrt{2})$

C $(\sqrt{2}, -9)$

D $(-9, \sqrt{2})$

Classified Practice

Complex Numbers

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{3}{4}, \frac{2}{5}\right)$

B $\left(\frac{2}{5}, -\frac{3}{4}\right)$

C $\left(\frac{7}{5}, -\frac{3}{4}\right)$

D $\left(\frac{2}{5}, \frac{1}{4}\right)$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, -\sqrt{11})$

B $(-\sqrt{11}, 0)$

C $(1 - \sqrt{11}, 0)$

D $(-\sqrt{11}, 1)$

Classified Practice

Complex Numbers

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0, 1 + \sqrt{13})$

B $(\sqrt{13}, 0)$

C $(0, \sqrt{13})$

D $(1, \sqrt{13})$

Q118 MCQ

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y) :

Choose the correct option.

A $(3, -1)$

B $(2, -2)$

C $(-1, 2)$

D $(2, -1)$

Classified Practice

Complex Numbers

Q119 MCQ

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

Choose the correct option.

A $(-1, 2)$

B $(0, 2)$

C $(-1, 1)$

D $(2, -1)$

Q120 MCQ

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

Choose the correct option.

A (3, 1)

B (1, 3)

C (2, 3)

D (1, 2)

Classified Practice

Complex Numbers

Q121 MCQ

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x,y):

Choose the correct option.

A (1, -2)

B (-2, 1)

C (-1, 1)

D (-2, 0)

Q122 MCQ

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x,y):

Choose the correct option.

A (3, -2)

B (-1, 3)

C (3, -1)

D (4, -1)

Classified Practice

Complex Numbers

Q123 MCQ

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x,y):

Choose the correct option.

A (2, -2)

B (1, -3)

C (-2, 1)

D (1, -2)

Q124 MCQ

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x,y):

Choose the correct option.

A (0, -2)

B (-1, -3)

C (-2, -1)

D (-1, -2)

Classified Practice

Complex Numbers

Q125 MCQ

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

Choose the correct option.

A (2, 1)

B 3

C (2, 2)

D (3, 2)

Q126 MCQ**For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:****Choose the correct option.**

A $1 + 6i$

B $-2 - 6i$

C $2 + 6i$

D $3 + 6i$

Classified Practice

Complex Numbers

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:

Choose the correct option.

A $2 - 2i$

B $-3 + 2i$

C $3 - 2i$

D $4 - 2i$

Q128 MCQ**For $(1 + 2i)(3 - i)$, select the simplified standard form:****Choose the correct option.**

A $6 + 5i$

B $4 + 5i$

C $-5 - 5i$

D $5 + 5i$

Classified Practice

Complex Numbers

Q129 MCQ

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

Choose the correct option.

- A 4
- B 5
- C 3
- D -4

Q130 MCQ

For $(2 - i)^2$, select the simplified standard form:

Choose the correct option.

A $2 - 4i$

B $-3 + 4i$

C $(2 - i)^2$

D $4 - 4i$

Classified Practice

Complex Numbers

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:**Choose the correct option.**

A $-8 + 6i$

B $-7 + 6i$

C $-9 + 6i$

D $8 - 6i$

Q132 MCQ**For i^{37} , select the simplified standard form:****Choose the correct option.****A** $-i$ **B** i **C** $1 + i$ **D** $-1 + i$

Classified Practice

Complex Numbers

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:

Choose the correct option.

A $1 - i$

B $-1 + i$

C i

D $-2 + i$

Q134 MCQ**For $i^{15} - i^6$, select the simplified standard form:****Choose the correct option.****A** $-1 + i$ **B** $1 - i$ **C** $2 - i$ **D** $-i$

Classified Practice

Complex Numbers

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:

Choose the correct option.

A $4 + 3i$

B $-5 - 3i$

C $5 + 3i$

D $6 + 3i$

Q136 MCQ**For $(2 + i)(2 - i)$, select the simplified standard form:****Choose the correct option.****A** 5**B** 6**C** 4**D** -5

Classified Practice

Complex Numbers

Q137 MCQ

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

Choose the correct option.

A $-3 + 4\sqrt{2}i$

B $2 - 4\sqrt{2}i$

C $(\sqrt{2} + 2i)^2$

D $-1 + 4\sqrt{2}i$

Q138 MCQ**For $i^8 + i^{11} + i^{14}$, select the simplified standard form:****Choose the correct option.****A $-1 - i$** **B i** **C $-i$** **D $1 - i$**

Classified Practice

Complex Numbers

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:**Choose the correct option.**

A $19 - 26i$

B $17 - 26i$

C $-18 + 26i$

D $(3 - i)^3$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:

Choose the correct option.

A $1 + 5i$

B $-1 + 5i$

C $-5i$

D $5i$

Classified Practice

Complex Numbers

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:

Choose the correct option.

A i

B $-2 + i$

C $1 - i$

D $-1 + i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Choose the correct option.

A $3 + 6i$

B $1 + 6i$

C $-2 - 6i$

D $2 + 6i$

Classified Practice

Complex Numbers

Q143 MCQ

For $(2 - 3i)^2$, select the simplified standard form:**Choose the correct option.**A $-6 - 12i$ B $5 + 12i$ C $(2 - 3i)^2$ D $-4 - 12i$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$., the required value is:

Choose the correct option.

A $4 + 2i$

B $3 - 2i$

C $3 + 2i$

D $-3 - 2i$

Classified Practice

Complex Numbers

Q145 MCQ

For the expression State the complex conjugate of $-4 + \sqrt{3}i$, the required value is:

Choose the correct option.

A $-4 - \sqrt{3}i$

B $4 + \sqrt{3}i$

C $-3 - \sqrt{3}i$

D $-4 + \sqrt{3}i$

Q146 MCQ**For the expression State the complex conjugate of $7i$., the required value is:****Choose the correct option.****A** $1 - 7i$ **B** 3 **C** $-7i$ **D** $7i$

Classified Practice

Complex Numbers

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$; the required value is:

Choose the correct option.

A $1 - \sqrt{5}$

B 3

C $-\sqrt{5}$

D $\sqrt{5}$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$, the required value is:

Choose the correct option.

A $\frac{1}{2} - 3i$

B $\frac{1}{2} + 3i$

C $-\frac{1}{2} - 3i$

D $\frac{3}{2} + 3i$

Classified Practice

Complex Numbers

Q149 MCQ

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{1}{2} + \frac{3i}{2}$

B $\frac{3}{2} + \frac{3i}{2}$

C $-\frac{1}{2} + \frac{3i}{2}$

D $-\frac{1}{2} - \frac{3i}{2}$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:

Choose the correct option.

A $1 - 2i$

B $2 - 2i$

C $-2i$

D $-1 + 2i$

Classified Practice

Complex Numbers

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:**Choose the correct option.**

A $-\frac{1}{10} - \frac{7i}{10}$

B $\frac{1}{10} + \frac{7i}{10}$

C $\frac{11}{10} + \frac{7i}{10}$

D $-\frac{9}{10} + \frac{7i}{10}$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-2 + 2i$

B $1 - 2i$

C $i(2 + i)$

D $2i$

Classified Practice

Complex Numbers

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:

Choose the correct option.

A $1 - i$

B $-1 - i$

C i

D $-i$

Q154 MCQ**For $1/i$, select the value in the form $a+bi$:****Choose the correct option.****A** $1 - i$ **B** $-1 - i$ **C** i **D** $-i$

Classified Practice

Complex Numbers

Q155 MCQ

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:**Choose the correct option.**

A $\frac{5}{13} - \frac{12i}{13}$

B $-\frac{5}{13} + \frac{12i}{13}$

C $\frac{8}{13} + \frac{12i}{13}$

D $-\frac{18}{13} + \frac{12i}{13}$

Q156 MCQ

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{-\sqrt{2} - i}{\sqrt{2} - i}$

B $\frac{\sqrt{2} + i}{\sqrt{2} - i}$

C $\frac{2\sqrt{2}}{\sqrt{2} - i}$

D $\frac{2i}{\sqrt{2} - i}$

Classified Practice

Complex Numbers

Q157 MCQ

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{8}{13} - \frac{14i}{13}$

B $-\frac{18}{13} - \frac{14i}{13}$

C $\frac{5}{13} + \frac{14i}{13}$

D $-\frac{5}{13} - \frac{14i}{13}$

Q158 MCQ

For $7i/(1 - i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-\frac{9}{2} + \frac{7i}{2}$

B $\frac{7}{2} - \frac{7i}{2}$

C $-\frac{7}{2} + \frac{7i}{2}$

D $-\frac{5}{2} + \frac{7i}{2}$

Classified Practice

Complex Numbers

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$, the required value is:

Choose the correct option.

A $-2 + 5i$

B $-2 - 5i$

C $2 + 5i$

D $-1 - 5i$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

Choose the correct option.

- A $-6 - \sqrt{7}i$
 - B $7 + \sqrt{7}i$
 - C $6 - \sqrt{7}i$
 - D $6 + \sqrt{7}i$
-
-
-

Classified Practice

Complex Numbers

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$; the required value is:

Choose the correct option.

- A 3
- B $\sqrt{2}i$
- C $-\sqrt{2}i$
- D $1 + \sqrt{2}i$

Q162 MCQ**For the expression State the complex conjugate of 9;, the required value is:****Choose the correct option.****A** 10**B** 3**C** 9**D** -9

Classified Practice

Complex Numbers

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$, the required value is:

Choose the correct option.

A $-\frac{3}{4} - 2i$

B $\frac{3}{4} + 2i$

C $\frac{1}{4} - 2i$

D $-\frac{3}{4} + 2i$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{6}{5} + \frac{12i}{5}$

B $-\frac{4}{5} + \frac{12i}{5}$

C $-\frac{1}{5} - \frac{12i}{5}$

D $\frac{1}{5} + \frac{12i}{5}$

Classified Practice

Complex Numbers

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:**Choose the correct option.**

A $\frac{7}{17} - \frac{6i}{17}$

B $\frac{24}{17} - \frac{6i}{17}$

C $-\frac{10}{17} - \frac{6i}{17}$

D $-\frac{7}{17} + \frac{6i}{17}$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{9}{13} + \frac{6i}{13}$

B $\frac{22}{13} + \frac{6i}{13}$

C $-\frac{4}{13} + \frac{6i}{13}$

D $-\frac{9}{13} - \frac{6i}{13}$

Classified Practice

Complex Numbers

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

Choose the correct option.

- A $-\frac{2i}{\sqrt{3} + i}$
- B $\frac{-\sqrt{3} + i}{\sqrt{3} + i}$
- C $\frac{\sqrt{3} - i}{\sqrt{3} + i}$
- D $\frac{2\sqrt{3}}{\sqrt{3} + i}$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $\frac{1}{2} - \frac{i}{2}$

B $\frac{3}{2} - \frac{i}{2}$

C $-\frac{1}{2} - \frac{i}{2}$

D $-\frac{1}{2} + \frac{i}{2}$

Classified Practice

Complex Numbers

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$., the required value is:

Choose the correct option.

A $-4 - 7i$

B $5 + 7i$

C $4 - 7i$

D $4 + 7i$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$., the required value is:

Choose the correct option.

- A $\sqrt{10} - 2i$
- B $-\sqrt{10} + 2i$
- C $1 + \sqrt{10} - 2i$
- D $\sqrt{10} + 2i$

Classified Practice

Complex Numbers

Q171 MCQ

For the expression State the complex conjugate of $-8i$., the required value is:

Choose the correct option.

A 3

B $8i$

C $-8i$

D $1 + 8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

Choose the correct option.

A $-\frac{5}{13} - \frac{12i}{13}$

B $\frac{5}{13} + \frac{12i}{13}$

C $\frac{18}{13} + \frac{12i}{13}$

D $-\frac{8}{13} + \frac{12i}{13}$

Classified Practice

Complex Numbers

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i :

Choose the correct option.

- A $5i$
- B $i\sqrt{5}$
- C $-i\sqrt{5}$
- D $\sqrt{5}$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i :

Choose the correct option.

- A $2i\sqrt{5}$
 - B $-2i\sqrt{5}$
 - C $4i\sqrt{5}$
 - D $i\sqrt{20}$
-
-
-

Classified Practice

Complex Numbers

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $-3i\sqrt{2}$

B $3i\sqrt{2}$

C $-6i$

D $-i\sqrt{18}$

Q176 MCQ

For $\sqrt{-8}$, select the simplified value in terms of i :

Choose the correct option.

A $2i\sqrt{2}$

B $\pm 8i$

C $\pm 2i\sqrt{2}$

D $\pm i\sqrt{8}$

Classified Practice

Complex Numbers

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i :

Choose the correct option.

A 3

B -3

C $3i$ D $\sqrt{9}i$

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i :

Choose the correct option.

A 2

B -4

C 4

D $-4i$

Classified Practice

Complex Numbers

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

- A -2
- B $i\sqrt{4}$
- C $-2i$
- D $2i$

Q180 MCQ

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

Choose the correct option.

- A $1 - 4i\sqrt{3}$
 - B $7 + 4i$
 - C $4 + 2i\sqrt{3}$
 - D $1 + 4i\sqrt{3}$
-
-
-

Classified Practice

Complex Numbers

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i :

Choose the correct option.

A $-6i\sqrt{2}$

B $12i$

C $i\sqrt{72}$

D $6i\sqrt{2}$

Q182 MCQ

For square roots of -50 , select the simplified value in terms of i :

Choose the correct option.

- A $\pm i\sqrt{50}$
 - B $5i\sqrt{2}$
 - C $\pm 10i$
 - D $\pm 5i\sqrt{2}$
-
-
-

Classified Practice

Complex Numbers

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

Choose the correct option.

A -6

B 6

C $-6i$

D 3

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

Choose the correct option.

A $2i$

B -2

C $-i\sqrt{2}$

D $-2i$

Classified Practice

Complex Numbers

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

Choose the correct option.

- A $6\sqrt{2}$
- B $6i\sqrt{2}$
- C $2i\sqrt{2}$
- D $-6i\sqrt{2}$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $4\sqrt{2}$

B $4i\sqrt{2}$

C $8i\sqrt{2}$

D $-4i\sqrt{2}$

Classified Practice

Complex Numbers

Q187 MCQ

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i :

Choose the correct option.

A $3 + 2i\sqrt{10}$

B $7 + 2i\sqrt{10}$

C $3 - 2i\sqrt{10}$

D $5 + 2i\sqrt{2}$

Q188 MCQ

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i :

Choose the correct option.

A 4

B $14i$

C $9 - 5i$

D 14

Classified Practice

Complex Numbers

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i :

Choose the correct option.

- A $7i$
- B $\sqrt{7}$
- C $i\sqrt{7}$
- D $-i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i :

Choose the correct option.

- A $2i\sqrt{7}$
- B $-2i\sqrt{7}$
- C $4i\sqrt{7}$
- D $i\sqrt{28}$

Classified Practice

Complex Numbers

Q191 MCQ

For $\sqrt{-12}$, select the simplified value in terms of i :

Choose the correct option.

A $\pm 2i\sqrt{3}$

B $\pm i\sqrt{12}$

C $2i\sqrt{3}$

D $\pm 4i$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

A $-3i$

B $3i$

C -3

D $i\sqrt{3}$

Classified Practice

Complex Numbers

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

- A 6
- B $-6i$
- C -3
- D -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

Choose the correct option.

- A -4
- B $-i\sqrt{4}$
- C $-4i$
- D $4i$

Classified Practice

Complex Numbers

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

Choose the correct option.

A $-5i\sqrt{2}$

B $5\sqrt{2}$

C $5i\sqrt{2}$

D $i\sqrt{26}$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

Choose the correct option.

- A $-1 + 2i\sqrt{2}$
- B $3 + 2i\sqrt{2}$
- C $-1 - 2i\sqrt{2}$
- D $1 + 2i$

Classified Practice

Complex Numbers

Q197 MCQ

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

Choose the correct option.

A $16 - 3i$

B 19

C 13

D $19i$

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i :

Choose the correct option.

A $-5i\sqrt{3}$

B $15i$

C $i\sqrt{75}$

D $5i\sqrt{3}$

Classified Practice

Complex Numbers

Q199 MCQ

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:**Choose the correct option.**

A $\left(-\frac{1}{2}, -7\right)$

B $\left(-\frac{1}{2}, 7\right)$

C $\left(-\frac{1}{2}, 6\right)$

D $\left(\frac{1}{2}, 7\right)$

Q200 MCQ

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:

Choose the correct option.

A $\{0, 1, 2\}$

B $\left\{2, \frac{1}{2}\right\}$

C $\left\{-2, 2, \frac{1}{2}\right\}$

D $\left\{0, 2, \frac{1}{2}\right\}$

Classified Practice

Complex Numbers

Q201 MCQ

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:**Choose the correct option.**

A $((1 + 7i)/5)^{2027}$

B $((7 + i)/5)^{2027}$

C $((7 - i)/5)^{2027}$

D $(7 + i)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x - 6)^2 = -8$:**Choose the correct option.**

A $x = 2 \pm 2i\sqrt{2}$

B $x = 2 \pm (2/3)i\sqrt{2}$

C $x = 2 \pm i\sqrt{2}$

D $x = 3 \pm i\sqrt{2}$

Classified Practice

Complex Numbers

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

For $4-3i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(4, -3)$

B $(4, 3)$

C $(-4, -3)$

D $(-3, 4)$

Quiz MCQ 2 Concept Quiz · MCQ

For $(1+2i)(2-i)$, select the simplified value:

Choose the correct option.

A $4 + 3i$

B 1

C 2

D $-4 - 3i$

Classified Practice

Complex Numbers

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

Choose the correct option.

A 1

B 2

C $-1 - 2i$

D $1 + 2i$

Quiz MCQ 4 **Concept Quiz · MCQ**

For $\sqrt{-27}$, select the expression in terms of i :

Choose the correct option.

- A $3i\sqrt{3}$
 - B $-3i\sqrt{3}$
 - C $9i$
 - D $i\sqrt{27}$
-
-
-

Classified Practice

Complex Numbers

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

Classified Practice

Complex Numbers

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Answer key

Use the question number shown on each page.

Q001 $(3, -2)$	Q027 $3 - 4i$	Q053 2	Q079 $3i\sqrt{3}$
Q002 $(4/3, -\sqrt{5}/3)$	Q028 -1	Q054 $8i$	Q080 $-2i\sqrt{3}$
Q003 $(-8, 0)$	Q029 $1 + 8i$	Q055 $1 - i$	Q081 $\pm 2i\sqrt{3}$
Q004 $(0, 2)$	Q030 $8 - i$	Q056 $-1 - i$	Q082 $-3\sqrt{2}$
Q005 $x = -1, y = 4$	Q031 0	Q057 $3 - 4i$	Q083 $-3i$
Q006 $(-3, 5)$	Q032 $4 - i$	Q058 $9 - 8i$	Q084 $7 + 6\sqrt{2}i$
Q007 $(-1/2, -\sqrt{3}/2)$	Q033 $1 - 3i$	Q059 $8 - i\sqrt{3}$	Q085 $i\sqrt{2}$
Q008 $(1, 0)$	Q034 -2	Q060 5	Q086 $3i$
Q009 $(0, 3)$	Q035 $-1 + 4\sqrt{2}i$	Q061 -15	Q087 $3i\sqrt{5}$
Q010 $x = 6, y = -2$	Q036 $7 + 11i$	Q062 $\sqrt{7}$	Q088 $-2i\sqrt{6}$
Q011 $x = -2, y = 3$	Q037 $1/15 - 13/14i$	Q063 $-i$	Q089 $-6i$
Q012 $(2, 5)$	Q038 $-3 - 4i$	Q064 $50i$	Q090 $\pm i\sqrt{2}$
Q013 $(-25, 13)$	Q039 -3	Q065 $-i\sqrt{6}$	Q091 $\pm 3i\sqrt{2}$
Q014 $(2/3, -\sqrt{5}/3)$	Q040 4	Q066 i	Q092 $\pm 4i$
Q015 $(7/2, -\sqrt{2}/2)$	Q041 i	Q067 $7/5 - 1/5i$	Q093 $-2\sqrt{3}$
Q016 $(-4, 0)$	Q042 -1	Q068 $2/3 + \sqrt{5}/3i$	Q094 $-14\sqrt{5}$
Q017 $(1/4, 0)$	Q043 $-i$	Q069 $2 + i$	Q095 $-2i$
Q018 $(0, \sqrt{3})$	Q044 $x = 0$ or $x = 1$	Q070 $3/5 + 4/5i$	Q096 $-(3/2)i$
Q019 $(0, -1)$	Q045 $2 - 3i$	Q071 $9/13 - 7/13i$	Q097 $13i\sqrt{2}$
Q020 $x = 3, y = 2$	Q046 $7 + i\sqrt{11}$	Q072 $-i$	Q098 $6i\sqrt{2}$
Q021 $x = 3, y = -3$	Q047 6	Q073 $1/3 - 2/3i$	Q099 $-2 + 2i\sqrt{15}$
Q022 $x = 2, y = -2$	Q048 $5i$	Q074 $((-2 - 144i)/85)^{2027}$	Q100 7
Q023 $x = 2, y = 1$	Q049 $3/5 + 4/5i$	Q075 $\pm i\sqrt{6}$	Q101 $x = 4 \pm (\sqrt{10}/2)i$
Q024 $x = 1/2, y = 3/2$	Q050 $-1/2 - 3/2i$	Q076 $-6\sqrt{2}$	Q102 C
Q025 $x = -1, y = 3$	Q051 $1 + 4i$	Q077 $-(2/3)i$	Q103 D
Q026 $1 - 7i$	Q052 $\sqrt{2} - i\sqrt{10}$	Q078 $i\sqrt{7}$	Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A

Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C

Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
Q173 B
Q174 A
Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B

Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$